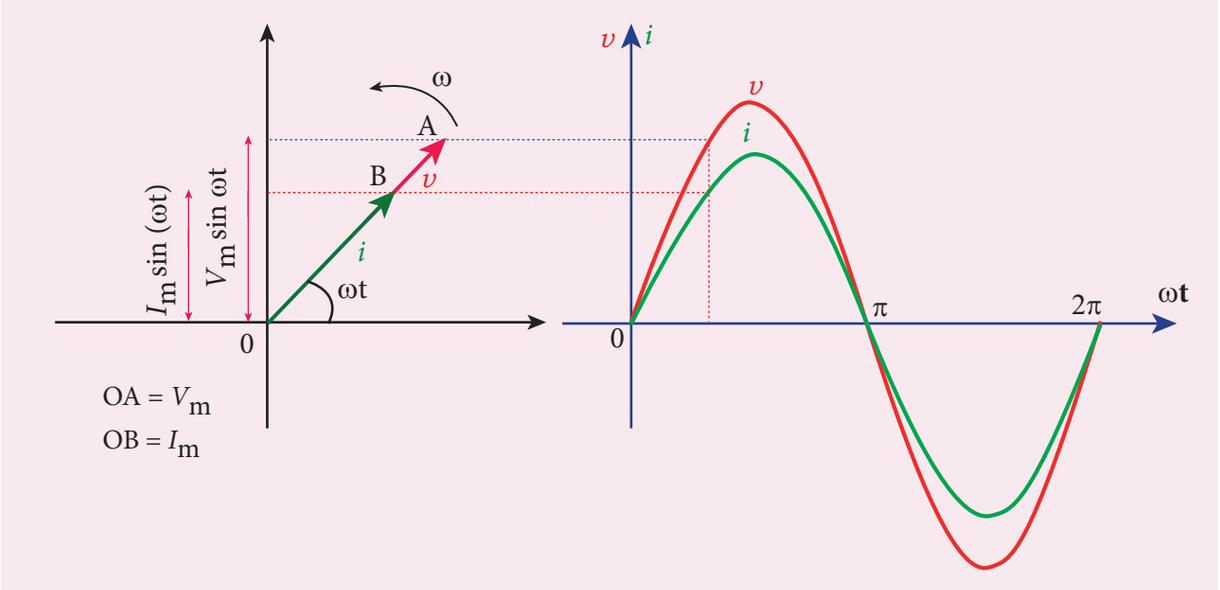


Figure 4.41 Phasor diagram and wave diagram for AC circuit with R

the applied voltage and the current are in phase with each other in a resistive circuit. It means that they reach their maxima and minima simultaneously. This is indicated in the phasor diagram (Figure 4.41). The wave diagram also depicts that current is in phase with the applied voltage (Figure 4.41).

4.7.4 AC circuit containing only an inductor

Consider a circuit containing a pure inductor of inductance L connected across an alternating voltage source (Figure 4.42). The instantaneous value of the alternating voltage is given by

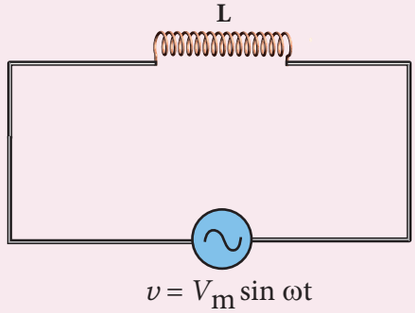


Figure 4.42 AC circuit with inductor

$$v = V_m \sin \omega t \qquad (4.40)$$

The alternating current flowing through the inductor induces a self-induced emf or back emf in the circuit. The back emf is given by

$$\text{Back emf}, \varepsilon = -L\frac{di}{dt}$$

By applying Kirchoff's loop rule to the purely inductive circuit, we get

$$v + \varepsilon = 0$$
$$V_m \sin \omega t = L\frac{di}{dt}$$
$$di = \frac{V_m}{L}\sin \omega t \; dt$$

Integrating both sides, we get

$$i = \frac{V_m}{L}\int \sin \omega t \, dt$$
$$i = \frac{V_m}{L\omega}(-\cos \omega t) + \text{constant}$$

The integration constant in the above equation is independent of time. Since the voltage in the circuit has only time dependent part, we can take the time independent part in the current (integration constant) as zero.

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